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Performance evaluation of a supercapacitor based synthetic inertia system using frequency locked loop and real time frequency derivative estimation

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ABSTRACT

The installation of intermittent renewable energy sources (RESs) are now fast increasing, reducing fossil fuel use. Due to the fact that RESs are typically grid-connected to the power-system via grid following fast-response voltage source converters, the amount of mechanical inertia synchronised in the power system is decreasing. As a result, the power system will be more sensitive to load and generation variations, causing larger frequency fluctuations which, in turn, could result in undesirable load-shedding, or large-scale blackouts. This paper presents a performance evaluation of a concept of emulating virtual inertia via power-electronics and a energy storage unit in terms of a supercapacitor connected to the DC-bus. Simulation and experimental results are presented to validate the proposed combination of frequency estimation and frequency derivative estimation. The paper shows the feasibility of using virtual inertia and how it could be implemented.

1. Introduction

The demand for clean and renewable energy in the Nordic power system is increasing due to the decommission of nuclear power and expected load increase. Traditionally, the inertial response in the Nordic power grid was supplied by large and heavy rotating synchronous generators in hydropower and nuclear power plants. The Nordic power system is moving towards a higher amount of wind and solar power in the future, which increases the amount of non directly connected generators. Most modern renewable energy sources (RES) are connected to the grid either via asynchronous generators or via power electronics, which do not provide a full inertial response to frequency changes unless it is synthesised in the converter. The integration of wind power in Sweden is estimated to increase over the coming years [1]. With fewer synchronous generators connected to the grid the total system inertia is expected to decrease from 300 GWs to 150 GWs by 2030 [2]. Reduced grid inertia will increase frequency oscillations in the power grid [3] which, in turn, will increase the mechanical wear and tear in power plants providing frequency control. In the Nordic grid these are mainly hydropower plants.

Due to a higher penetration of loads which are connected via power electronics to the grid, the amount of frequency and voltage-dependent loads will also be reduced. This must be taken into consideration due to its reduced damping effect [4] during larger power unbalances. In order to keep the grid frequency within limits, faster frequency control measures must be implemented. The control strategies for fast frequency response can be divided into 4 groups [5]: emulated inertia control ($P\frac{df}{dt}$), Automatic Generation Control (AGC), $P\Delta f$ droop control or static trigged response. Recently a faster frequency control measure has started. It is called fast frequency response (FFR) and is required to go from zero to full active power below a certain frequency threshold in no less than 0.7 s, and not more than 1.3 s depending on the reserve participated in [6]. It can either consist of a grid-connected energy storage device which can increase its power output rapidly, or a controllable load which can be reduced or disconnected. The FFR will help to stabilise the system before the frequency containment reserve (FCR) starts acting. In the Nordic power system FCR is made up of mainly hydropower. An AGC system is based on a centralised controller that performs frequency regulation. An inertia emulation concept in AGC systems has been evaluated for a connected inverter where the DC-link voltage is modified to get a emulated power response during frequency disturbances [7]. $P\Delta f$ droop control is a dynamic frequency response and used in most conventional power systems. The $P\Delta f$ is based on a proportional power output of the generator to the deviation in frequency. In the Nordic grid, it is considered as FCR-N and is active between 49.9 Hz to 50.1 Hz.

The idea to create virtual inertia using power electronics and an energy storage (ES) system has been widely discussed [8]. The most common approach is the virtual synchronous generator (VSG) and virtual synchronous machine (VSM or VISMA) [9]. The VSG can be implemented in most power electronic controlled generation units, such as wind power. The risk with implementing a VSG into a wind power

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generator is that, since the energy is taken from the rotating parts, the turbine will not work at its Maximum Power Point (MPP) which can reduce the total power output of the turbine. When the turbine has to accelerate to its optimal rotational speed after a higher power output, the output power of the generator will decrease, this can cause a second dip in the frequency and thereby worsen the situation. GE's wind inertia system can increase its power output over a few seconds by 5%–10% during a larger frequency disturbance [10]. Typically these implementations are denoted as virtual or synthetic inertia but its power is not proportional to the derivative of the frequency as real inertia is. However, inertia response techniques and recovery techniques have been discussed [11,12]. The idea of using the energy stored in the capacitors on the DC-link has recently attracted focus, and linking the DC-link voltage with the frequency for emulating inertia has been suggested [13]. The similarity between the rotor speed in a SG and the varying DC-link voltage when emulating inertia has been studied. It has been shown that the DC-link voltage is proportional to the grid frequency in cases where the DC-link voltage is adjustable. This is not valid if the energy storage is large, for example in battery storage systems or large supercapacitor systems. One option is to emulate the inertia in High Voltage Direct Current (HVDC) connections which have a certain capacitor size, and where the DC-link voltage can be adjusted [14–16]. On the other hand, the same idea could be used to create a standalone system consisting of power electronics and an energy storage device. A standalone system would give a higher flexibility regarding the injection of SI power into the power system. One option could be to use electric vehicles to provide inertia to the grid while charging or discharging by implementing a VSG controller into the charger which has been studied in [17].

If the ES is too large and the DC-link voltage level cannot be adjusted, the derivative of the frequency can be calculated with an frequency locked loop (FLL) [18], where noise and voltage harmonics are not treated. One way to handle harmonics and noise in the frequency derivative estimation is by using a band-pass filter on the output from the FLL, which has been studied in [19].

Studies on using batteries as ES and PLL have been done using a Hardware In the Loop (HIL) system where it was shown that the ROCOF and minimum frequency nadir was improved in a micro-grid, however the derivative calculation was left for future studies [20].

Previous studies on synthetic inertia have compared different implementations, but voltage harmonics and noise have not been take into consideration while estimating the grid frequency and frequency derivative in real-time.

In this paper, we present a complete synthetic inertia (SI) system developed and experimentally tested based on a FLL frequency estimation combined with a Savitzky Golay Finite Impulse Response (FIR) filter for frequency derivative estimation. The paper starts with quantifying the need for inertia in the power grid in Section 2, and then explains the new method of calculating the grid frequency derivative with focus on harmonics and noise levels in Section 3. Thereafter, the proposed control-system and hardware are presented. In section IX the results from the hardware tests are compared with Matlab simulations.

2. Method

This section starts by describing the inertial response of the system compared to a synchronous machine. In the second part, trade-offs for the hardware and which components have been used in the system are defined.

2.1. Grid inertia

To quantify the need of synthetic inertia, we need to understand the dynamics of the synchronous machine which describes the system frequency. The designed system is intended to provide SI in normal operation, i.e. not saving the power system in an N-1 event.

The relation between the power balance on the grid and the system inertia is described by the swing equation

$$P_{gen} - P_{load} = \sum_i J_i \omega_{m,i} \frac{d\omega_{m,i}}{dt}, \tag{1}$$

where J_i is the total moment of inertia of the generator in kgm^2, $\omega_{m,i}$ is the angular speed of unit i, P_{gen} is generated power and P_{load} is the total electrical load on the grid. Here, we neglect damping and other non-linear phenomena.

During a change in electrical frequency, each synchronous generators will respond with an inertial response given by

$$P_{inertia} = J_i \left(\frac{2\pi}{n_P}\right)^2 f_{el} \frac{df_{el}}{dt}, \tag{2}$$

where n_P is the number of pole pairs, and neglecting rotor-angle oscillations.

The response from a synthetic inertia should behave in the same fashion and we define the synthetic inertia as

$$P_{SI} = K_{SI} \frac{df_{el}}{dt}, \tag{3}$$

where K_{SI} is the amount of inertia delivered at a certain frequency derivative. K_{SI} is limited by two factors, the maximum power rating and the amount of energy storage for a certain duration. The hardware limits the maximum power at the maximum expected frequency derivative $K_{SI} \leq \frac{P_{Max}}{\frac{df_{el}}{dt}}$. For normal operation in the power grid an maximum expected frequency derivative is $35mHz/s$ and P_{Max} is dependent on the hardware system.

The amount of stored energy limits the maximum power output depending on the expected maximum energy needed during a response which is left for future studies but it will depend on the desired duration of the SI response.

2.2. Hardware system considerations

The proposed electrical system for the grid connection of the synthetic inertia is presented in Fig. 1. The trade-offs for the hardware system were based on a micro grid it was connected to which is described in detail in [21]. The micro grid has a rated voltage of 400 V_{LL} with a maximum power of 60 kVA from the Synchronous generator. Due to the choice of grid voltage, standard components could be used for the lab setup but with a restriction of DC-bus voltage level. In normal operation of the grid we expect the charge discharge cycle to be about 1 Hz meaning the system should cope with a substantial amount of cycles with high power levels. This speaks in favour of a supercapacitor instead of a battery due to, no chemical reactions are required to charge or discharge the supercapacitor. Since there is no chemical reaction ongoing, the charge–discharge cycle life is much higher compared to, e.g. lithium-ion batteries [22]. The disadvantage being low voltage capability per module, low energy density, and high price. The SI system is intended to be activated during normal grid fluctuations and not as a saver for larger N-1 contingencies, where more power is needed. The energy storage needed for the inertia response consists of a 125 V 68 F supercapacitor supplied by Maxwell technologies, which can be fully cycled up to 1 000 000 times with high discharge and charge currents. The energy storage is directly connected to a 2-level Semikron IGBT based voltage source converter rated 1200 V and 250 A. After the converter, a harmonic filter of LC-topology is connected to reduce the total harmonic distortion (THD) and reduce the stress on components. Since the LC filter is connected to a ΔY transformer, the inductance of the transformer will form an LCL filter to the grid. Design considerations and performance of the LCL filter are discussed in [23,24]. The ΔY transformer is used to electrically isolate the system from the grid and thereby reducing fault currents and step-up voltage to the grid voltage. The transformer is ΔY connected to remove any zero sequence currents and third order voltage harmonics.

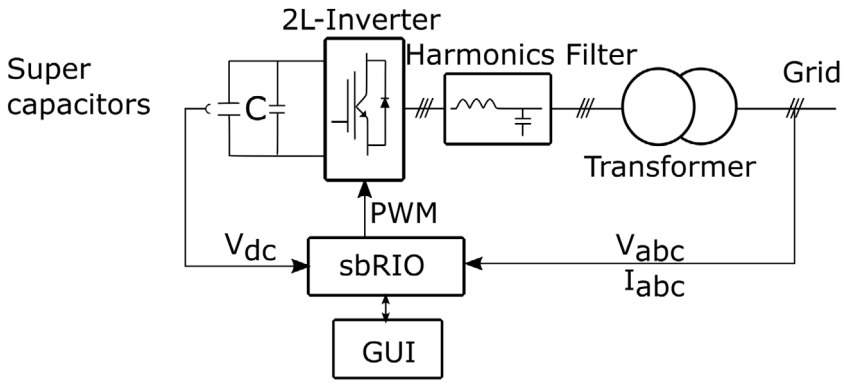


Fig. 1. Shows the proposed synthetic inertia system.

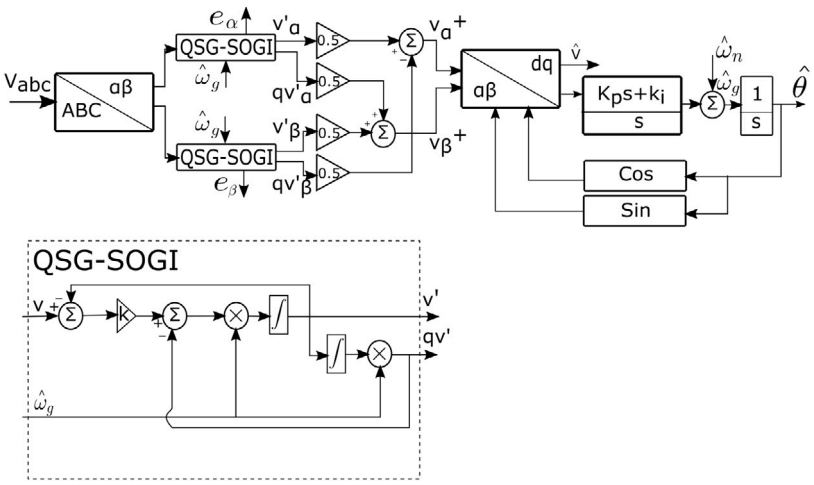


Fig. 2. Shows the structure of the QSG-DSOGI-PLL used to estimate the grid voltages and phase which consists of a Clark transform, two QSG-SOGI, park transform and a synchronous reference frame (SRF) PI loop.

The controller is based on a National instruments single board RIO (sbRIO) 9607, with a general-purpose inverter controller card (GPIC) 9684. The controller contains a Field Programmable Gate Array (FPGA) which has a clock frequency of 40MHz and a real-time system, which handles data transfer and communication.

As a comparison, the maximum equivalent inertia the used hardware is capable of emulating corresponds to a Hydropower generator in the 30 MVA to 50 MVA range, with a rotating weight of between 60 tons to 200 tons. An exact number cannot be given since the inertial response also depends on the rotational speed of the unit.

3. Grid estimation

Due to the inertia acting on the derivative of the electrical frequency, filtering of the electrical frequency is essential, since, for a derivative component in the regulator, noise is detrimental. A Frequency Locked Loop (FLL) was implemented in Simulink and in the FPGA hardware to track the grid frequency which minimises the frequency estimation error. Due to the sub-optimal phase tracking in the FLL a phase-locked loop (PLL) was used which minimises the error on the phase estimation for the grid connection of the system.

3.0.1. PLL

One common way to detect the voltage phase angle, amplitude, and frequency is to use a standard PLL [25–27]. A known drawback is that it is sensitive to noise and harmonics of the measured signals. The Quadrature Signal Generator-Double Second Order General Integral-PLL (QSG-DSOGI-PLL) seen in Fig. 2, is a robust PLL during large phase angle disturbances and has a fast response time. The QSG-DSOGI acts as a sinusoidal signal integrator and behaves as a band-pass filter due to its feedback of the output signal as seen in Fig. 2. The DSOGI-PLL in this paper is used to estimate the phase and voltage amplitude for connecting the energy storage to the electrical grid. The continuous transfer function of the QSG-DSOGI, is given by Eqs. (4b) and (4c). The band-pass and low-pass filter bandwidths are defined by k, where the settling time can be approximated by Eq. (4a). Eq. (4d) shows the continuous time open-loop transfer function, G(s), using a small signal assumption.

$$t_{s(SOGI)} \approx \frac{10}{k\omega'} \tag{4a}$$

$$\frac{v'(s)}{v(s)} = \frac{k\omega s}{s^2 + k\omega s + \omega^2} \tag{4b}$$

$$\frac{vq(s)}{v(s)} = \frac{k\omega^2}{s^2 + k\omega s + \omega^2} \tag{4c}$$

$$G(s) = V\hat{K}_p\left(\frac{1}{s\frac{Ts}{2} + 1}\right)\frac{1}{s} \tag{4d}$$

The output of the QSG-DSOGI is then used to extract the fundamental positive and negative voltage sequence (PNSC) where the positive is fed into the $\alpha\beta$ to dq transform. The dq transformation from the rotating reference frame values are then sent to the Synchronous Reference Frame (SRF) which is used to estimate the grid phase. The SRF frequency estimator is based on a PI controller.

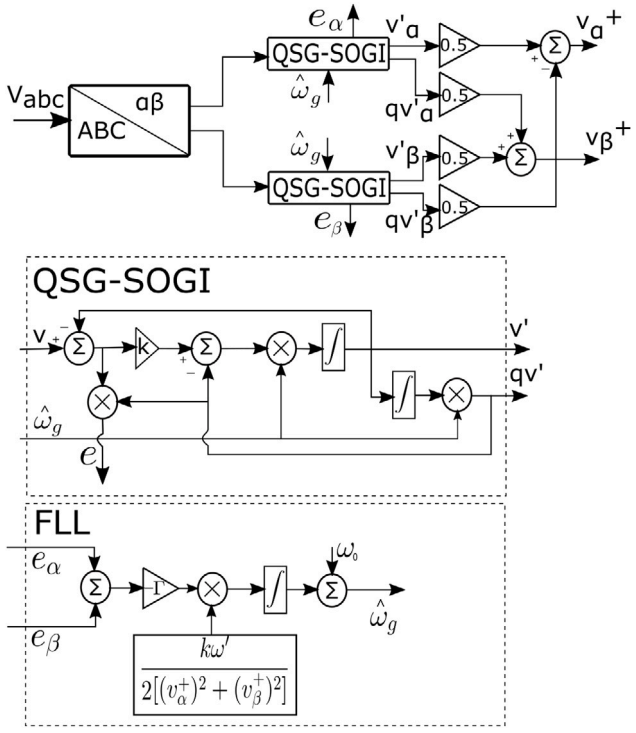


Fig. 3. Shows the structure of the DSOGI-FLL for the frequency estimation, consisting of a Clarke transformation in the top, SOGI filter for α and β in the middle, and the frequency-locked loop at the bottom.

3.0.2. FLL

To estimate the electrical grid frequency a frequency locked loop (FLL) is used which minimises the error on the frequency estimation. This issue has been widely researched [28–31]. The FLL is based on a QSG-DSOGI pre-filter where the transfer function is given by (4b) and (4c) and the bandwidth of the filter is set by K. The output of the SOGI is then fed into the PNSC to extract the positive voltage sequence which is used by the FLL to estimate the frequency. The structure of the FLL is shown in Fig. 3 where the settling time $t_{s(FLL)}$ can be approximated by Γ,

$$t_{s(FLL)} \approx \frac{5}{\Gamma}. \tag{5}$$

The ideal FLL transfer function will have a frequency response as shown in Fig. 4.

3.1. Derivative estimation

Estimating the derivative in real-time on a sampled signal containing noise is well known to be problematic. In this paper a moving polynomial fitting is used. It is implemented in a Finite Impulse Response Filter (FIR-Filter) to smooth the data, i.e. reduce the noise content, and estimate the frequency derivative. The polynomial fitting is applied and evaluated at the centre data point in the window. Savitzky and Golay (SG) found that this sliding window could be interpreted as a Linear Time Invariant system (LTI-System) which makes it possible to interpret the signal in the frequency domain instead [32,33]. This can be seen as a low-pass filter and the properties of the filter can be evaluated in the frequency domain. Looking at the grid frequency during a short time period we can linearise the swing equation and thereby catch the dynamics of the frequency [34]. Since the polynomial smoothing is a LTI system, a FIR filter can be used to implement the smoothing filter. The FIR filter can be described in frequency domain by the convolution theorem as

$$F(x * h) = F(x) \cdot F(h), \tag{6}$$

where $F(x)$ is the estimated grid frequency signal and $F(h)$ is the SG filter function. The FIR filter impulse response is described by

$$h[n] = \sum_{i=0}^{N} a_n \cdot \delta[n-i], \tag{7}$$

where a_n are the coefficients for the FIR filter which are pre-calculated and stored in a lookup table (LUT).

If the degree of the running least square polynomial is chosen correctly, and the polynomial can be differentiated, the convolution theorem gives that the resulting convoluted signal can be differentiated.

The design of the polynomial filter coefficients has been studied in [35] where the solution for the filter coefficients can be written as

$$a = (A^T A)^{-1} A^T x, \tag{8}$$

where a is the coefficient matrix, A is the design matrix for the polynomial approximation and x is the input signal vector. The SG filter introduce a phase delay which will be linear in the number of FIR taps and can be estimated when the sampling frequency, f_s, is known.

By fitting a low degree polynomial to sub-sets of the signal, the filter will give an estimate of the smoothed signal or the derivative of the smoothed signal at the centre point. The cutoff frequency is defined by the degree of the polynomial and the window length where the degree will mainly change the cutoff frequency while the window width will both change the cutoff frequency and the slope of the filter response.

3.2. Control system

3.2.1. Measurement system

The DC-bus voltage is measured by the inverter and sent as a 0 V to 10 V signal to the sbRIO Digital to Analogue Converter (DAC). The grid phase voltages and currents are measured after the transformer via an in-house designed three phase measurement card. The phase currents are measured via three LEM-HAIS50 Hall effect current sensors whose voltage signal is then converted with a Texas instruments XTR115U to a 4 mA to 20 mA current signal which is sent to the sbRIO. The current measurement range was ±50 A. The voltage sensors used in this setup were based on the isolated Broadcom ACPLC79B amplifier which then was converted to 4 mA to 20 mA current signal with a XTR115U chip and sent to the sbRIO. All of the signals are then collected on the interface card and converted to ±10 V which then are sampled by the DAC on the sbRIO GPIC card.

3.2.2. Frequency response

To estimate the frequency response of the system, ideal three-phase sinusoidal voltages were generated in Labview which replaced the real measured voltages. The ideal voltages replaced the measured three-phase voltages, since they could be controlled in both frequency and amplitude. A sinus in sinus out measurement was done where a fundamental 50 Hz signal was used which was overlaid with a sinusoidal signal of varying frequency, to get the open transfer function of the system.

$$V_a = V_a^0 \sin\left\{2\pi\left[50 + A_{\tilde{f}} \sin(2\pi\tilde{f})\right]\right\}. \tag{9}$$

Where the signal amplitude $A_{\tilde{f}}$ and frequency $\tilde{f}$ could be set.

3.2.3. Grid side control

The grid side phase voltages are measured and fed into a Double Second-Order generalised Integrator Phase Locked loop (DSOGI-PLL) which tracks the grid voltage vector and generates the grid phase angle Θ. The DSOGI-PLL is described in detail in Section 3.0.1 and used for the Voltage Oriented Control (VOC). The grid angle is then used to transform the voltages v_a, v_b, v_c and currents i_a, i_b, i_c from the stationary reference frame to the rotating synchronous reference frame v_d, v_q and i_d, i_q which will be DC components. To control the currents two PI regulators and one PD regulator are used. The two PI controllers are

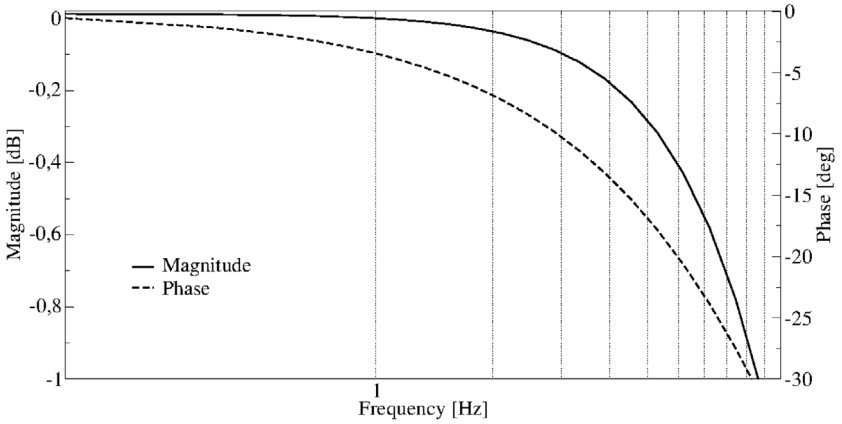


Fig. 4. Shows the Bode plot of the transfer function of the frequency locked loop shown in Fig. 3.

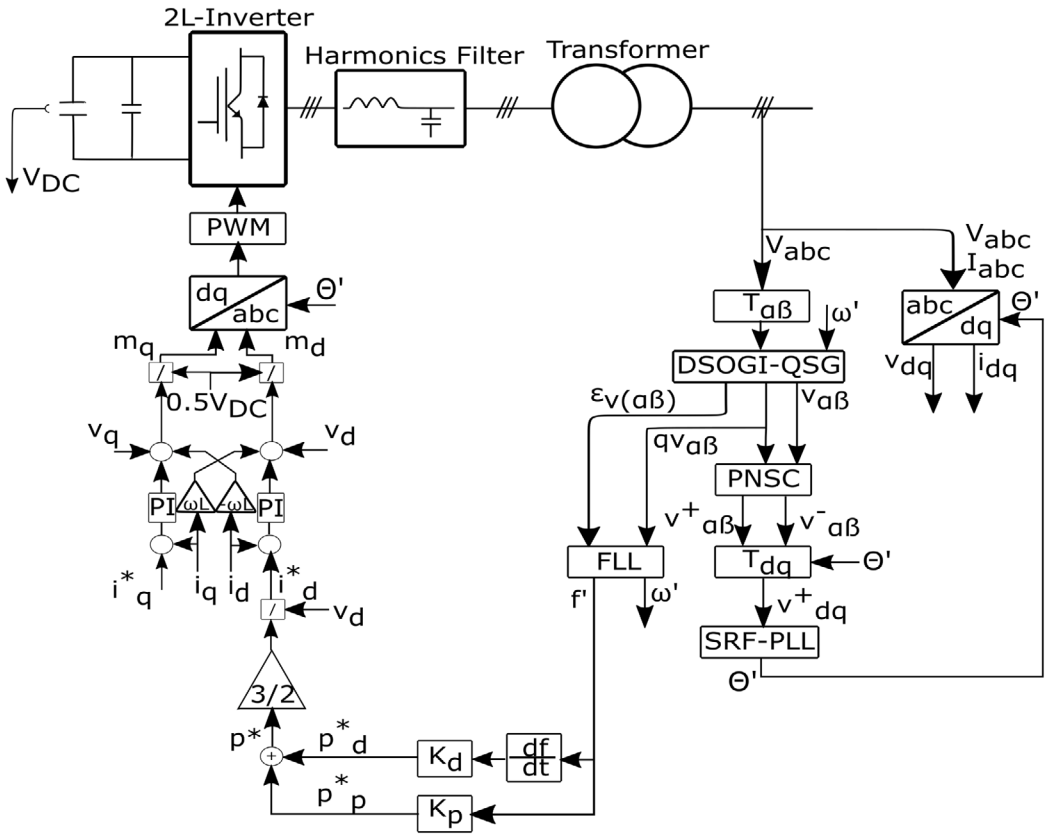


Fig. 5. Shows the proposed synthetic inertia control system topology. On the right side the grid phase and the frequency estimation are calculated from the grid voltage. In the mid bottom part the frequency derivative is estimation from the FLL frequency and the FCR controller are shown. The left side shows the sum of the power set points from the PD controller is converted to PWM signals.

inner control loops which control the d and q axis currents. The PD controller is used to emulate the inertia and FCR response of the system and gives the reference power P_d^* and P_p^*. The d-axis in the synchronous frame is aligned with the grid voltage vector to be able to apply the VOC controller, where the d-axis vector is set to be equal to the grid voltage vector. Since the system is only providing active power, the q-axis voltage is set to zero which represents the reactive power flow to the grid. The reactive and active power can be calculated as in Eqs. (10) and (11) when $V_q = 0$,

$$P_g = \frac{3}{2}(v_d i_d + v_q i_q) = \frac{3}{2} v_d i_d \tag{10}$$

$$Q_g = \frac{3}{2}(v_q i_d + v_d i_q) = \frac{3}{2} v_d i_q \tag{11}$$

The d and q axis reference currents can be calculated from Eqs. (10) and (11). The reactive power reference is set to zero. In this case only the d axis current reference needs to be calculated which results in Eq. (12).

$$i_d^* = \frac{2}{3}\frac{P_g^*}{v_d} \tag{12}$$

The reference currents are then compared with the real d and q currents as seen in Fig. 5 and the error is sent to a PI controller respectively. The control signals from the PI regulators are then added to the d and q voltages together with the decoupling terms before being normalised with the DC voltage as base. The reference voltages are then transformed back to the stationary reference frame and sent



Fig. 6. Shows the proposed hardware system with the numbered components 1:Control board, 2: Local power supply, 3: Transformer, 4: Filter, 5: Inverter, 6: Supercapacitor energy storage.

into the PWM block. A carrier-based Sine-PWM is used to generate the grid-following inverter pulses.

4. Results

This section will show the hardware system used for the measurements and the resulting measured dynamics of the proposed SI control system. Experimental results are also compared with a simulated response implemented in Matlab-Simulink. The system dynamics was tested with the cabinet shown in Fig. 6, which is a hardware implementation of the system shown in Fig. 5, connected to the local 400 V grid, and described above. The rated power of the system was 15 kVA, limited by the maximum current of the converter. The control system had a sampling frequency of 40 kHz which was sampled down via a 4 point moving average filter to 10 kHz to reduce the noise of the sampled signals. After the moving average filter, all data processing was done with a frequency of 10 kHz. The PLL was using $K = 1/\sqrt{2}$ for the DSOGI prefilter and $Ki = 2$, $Kp = 0.2$ for the SRF loop, which gave a sufficient fast response time but with a good noise rejection. The FLL was using $K = \sqrt{2}$ and $\Gamma = -50$ which gave a settling time of 50 ms. The settling time was a trade-off between the noise rejection capability of the FLL and the extra phase delay it will introduce to the total system. The SG filter used a second-order curve-fitting with a window width of 1800 taps and a loop frequency of 2 kHz. This turned out to be reasonable on the very stable Nordic power system where the frequency fluctuations during normal operation are quite small. For a low-inertia system the frequency derivative estimation can be done faster. However, a certain signal amplitude is needed to extract the derivative of the frequency. The decision to use a second-order curve with 1800 taps and 2 kHz loop frequency was a trade off between total phase delay and noise rejection. A setting, which enabled managing the normal Nordic grid frequency events which have a frequency derivative of the order of 35 mHz/s, was used. During the grid-connected tests, the supercapacitor were kept with a state of charge of around 90%. A slow SOC controller could be added to keep the system capable of providing virtual inertia and damping during longer periods to account for self-discharging.

The overlaid frequency signal in Eq. (9) was changed from 0.25 Hz to 5 Hz and with an amplitude decreasing from 0.6 Hz to 0.4 Hz not to saturate the frequency derivative estimation. The amplitude and phase shift of the FLL, frequency derivative, and power output were measured. The results of the frequency response of the separate control systems and the total system phase delay and damping can be seen in the Bode plots in Figs. 7 and 8.

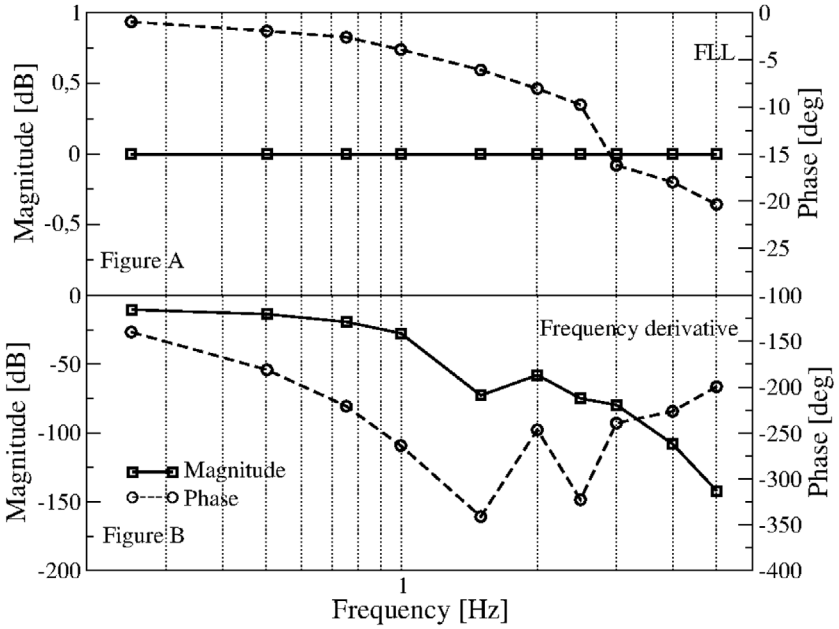


Fig. 7. Figure A shows the frequency response of the FLL and Figure B the response of the derivative where an ideal sinus was used as input and the phase shift and amplitude was measured on the output.

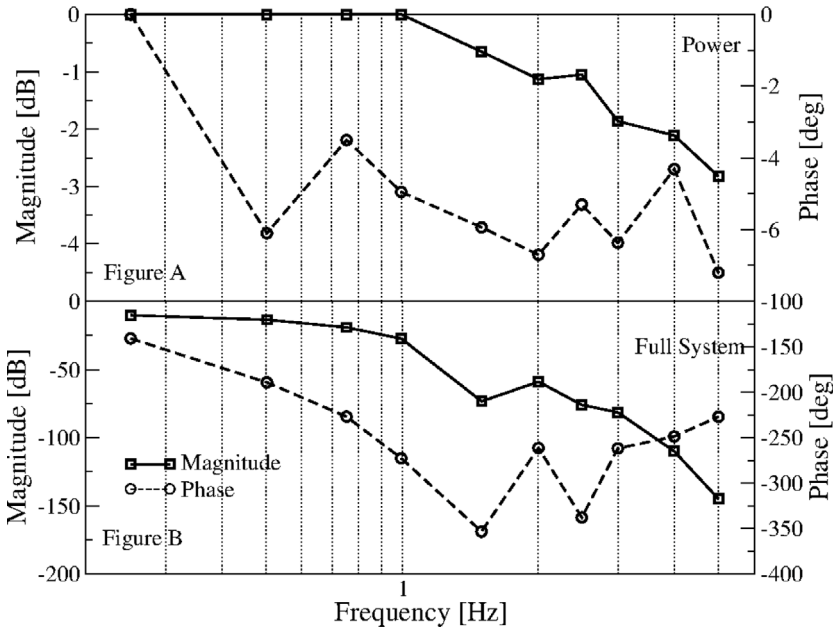


Fig. 8. Figure A shows the frequency response from the reference power set-point to the inverter output power where the reference power is given by the inertia response. Figure B shows the Bode plot of the transfer function from a three phase voltage reference to the inverter power output. Ideal sinuses were used as input and the phase shift and amplitude was measured.

Fig. 7A shows that the amplitude does not change from the input three-phase voltage to the frequency estimation of the FLL. The phase shift is smaller than 20 degrees up to 5 Hz which is expected when compared with the ideal bode plot in Fig. 4. Fig. 7B shows the frequency response of the derivative estimation done by the SG filter. It can be seen that the filter has a small damping at 0.25 Hz which increase to over 50 dB damping at 5 Hz. The dips in Fig. 7B at 2 Hz and 3 Hz are due to the taps of the FIR. Due to few samples, the taps of the filter displays sharp corners but the large phase shift is expected due to the linear relationship between the number of taps and sampling frequency of the FIR filter.

In Fig. 8A the response from power set point to output power is measured, where it can be seen that it dampens the signal after 1 Hz and has a low phase shift due to the fast PI current controller. Fig. 8B shows the total system frequency response from an input change in voltage frequency to output power to the grid where it can be seen that the derivative estimation is causing the largest phase shift and damping in the system.

The next step was to use the actual voltages from the grid. Figs. 9 and 10 shows the Fast Fourier Transforms (FFTs) that were performed on the three-phase voltages(A), FLL frequency estimation(B), filtered FLL frequency(C) and frequency derivative estimation(D) to analyse the

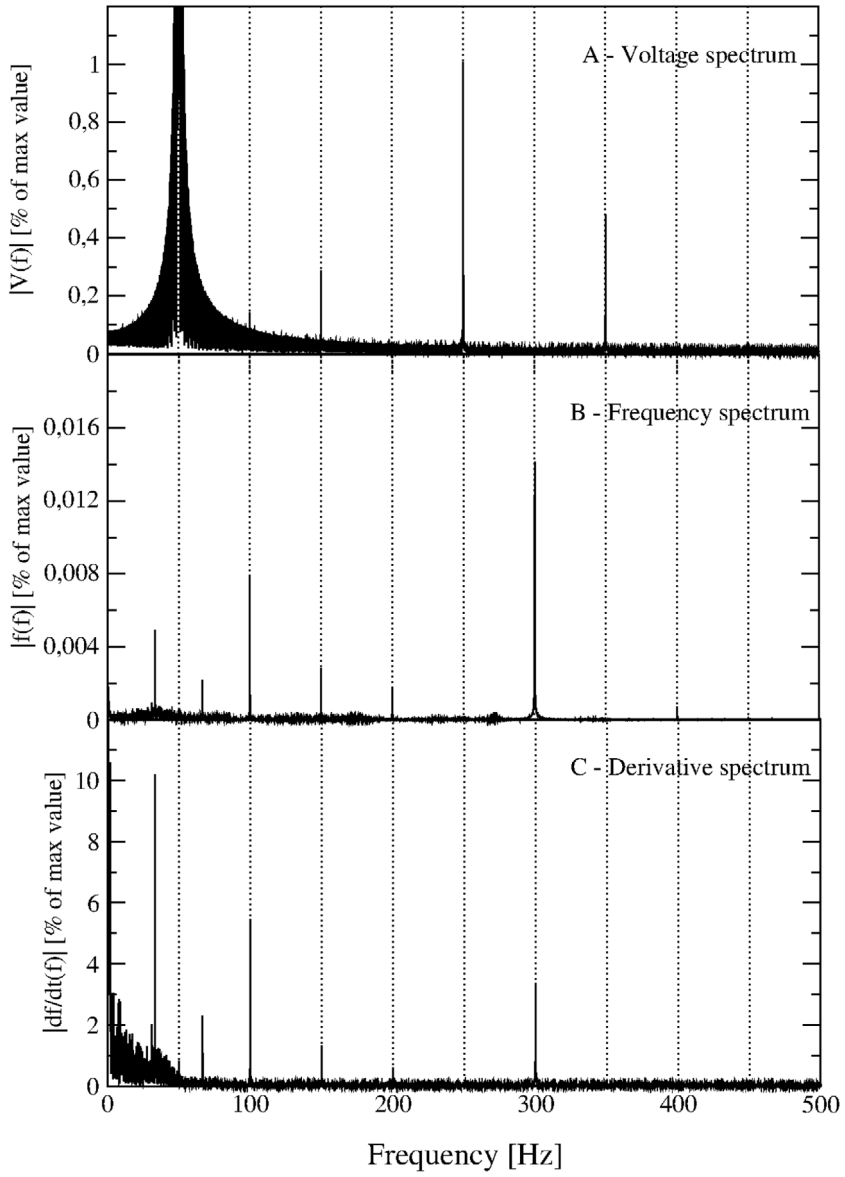


Fig. 9. Figure A shows the frequency spectrum of the three phase grid voltage. Figure B shows the frequency spectrum of the estimated grid frequency from the FLL. Figure C shows the frequency spectrum of the estimated grid frequency derivative with no LP filter where the frequency and derivative estimations are done in real-time on the FPGA. All signals are normalised to the largest harmonic for comparison.

harmonic content, while the system was grid-connected. In Fig. 9 the FFTs of the sampled signal without any extra filtering are shown.

In the voltage measurement the fundamental 50 Hz is clearly seen as well as the 3rd, 5th and 7th harmonics. Fig. 9 C displays larger peaks at 33.3 Hz, 66.6 Hz and 100 Hz, they are also found in the FLL spectrum. To be able to compare the signal amplitude of each harmonic the FFT is normalised and shown in percentage. Fig. 9A shows that the higher harmonics are small compared to the fundamental 50 Hz. In Fig. 9B it can be seen that the FLL introduces some low amplitude harmonics into the frequency estimate. The same frequency harmonics appear in Fig. 9C, the spectrum of the estimated frequency derivative. Since the harmonics are much higher than the power system oscillations which are lower in frequency, they can filtered away, especially for the relatively stable Nordic power system.

In Fig. 10A 15 Hz low pass filter is implemented after the frequency estimation of the FFL, which reduces the high-frequency noise in the extracted electrical frequency and thereby gives a better signal to noise ratio (SNR) on the derivative estimation.

The 15 Hz cutoff frequency was chosen since most noise contents were found above 15 Hz, and to minimise the extra phase delay on the derivative caused by the filter to the low frequencies which are wanted.

To estimate the total time delay introduced by the FLL and the derivative estimation, a step-change in frequency was performed in three ideal 50 Hz sinusoidal voltages generated in Labview which are then changed in a step-like manner from 50 Hz to 49.6 Hz. The voltages were saved and used in Matlab Simulink where the complete control system was simulated for comparison with the real-time estimation. Fig. 11A shows the system performance after a step change in frequency. Both the simulated response of the FLL and the real-time estimation match accordingly, also the real-time phase delay of the low-pass filter can be seen. The Matlab Simulink estimation and the real time Labview implementation coincide and the lines are on top of each other. In Fig. 11 B, results in the derivative from a step change in frequency can be seen, where the FPGA loop frequency was changed from 10 kHz to 1.6 kHz with a fixed window size of 1800 taps. The figure contains results from the simulated system in Matlab/Simulink and the real-time system implemented in the hardware.

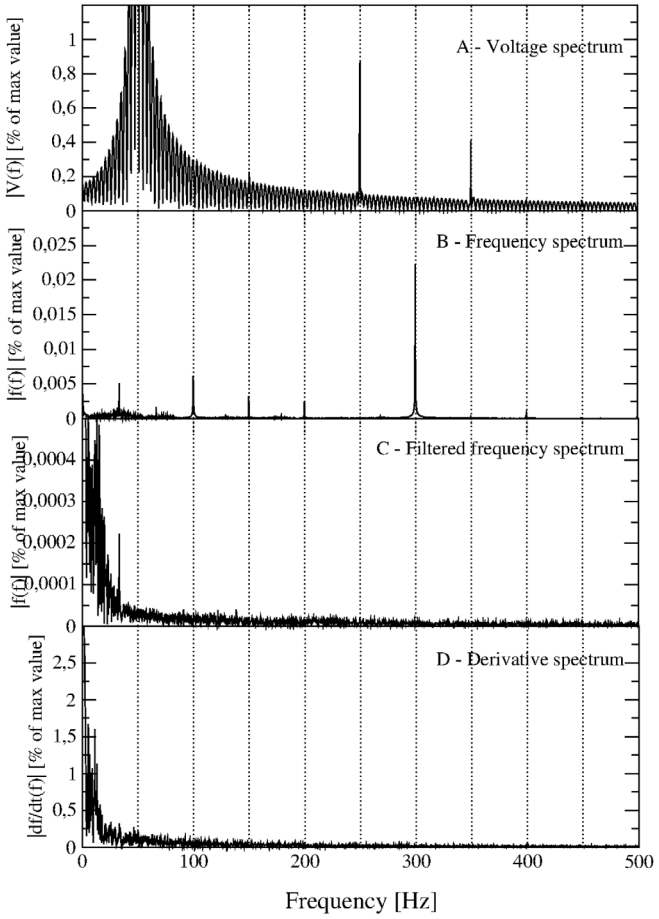


Fig. 10. Figure A shows the frequency spectrum of the voltage. Figure B and C show the frequency spectrum of the estimated frequency from the FLL before and after the low pass filter with a cutoff frequency at 15 Hz. Figure D shows the frequency spectrum of the estimated frequency derivative. The grid frequency and derivative is estimated in real-time on the FPGA.

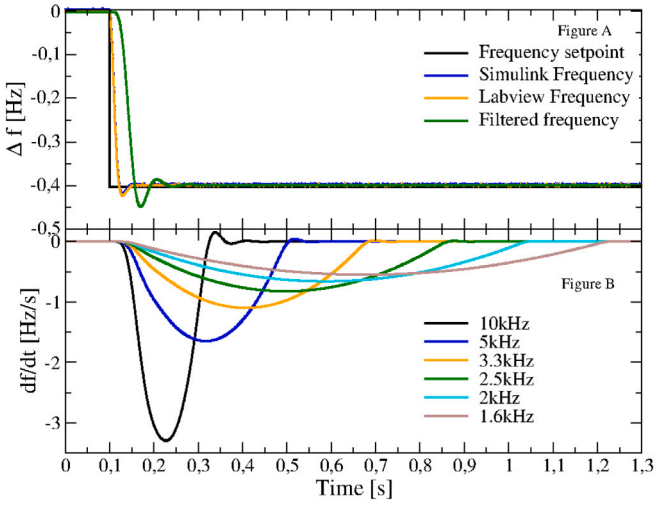


Fig. 11. Figure A shows the generated frequency step in the measured three phase voltage, the simulated frequency estimation and the real time estimated frequency from the FPGA. Figure B shows the simulated derivative estimation with sampling frequencies from 10 kHz down to 1.6 kHz.

It can be seen in Fig. 11 B that a slower loop frequency results in a smother signal, which can be compared with extending the number of taps and thereby decreasing the cutoff frequency of the FIR filter.

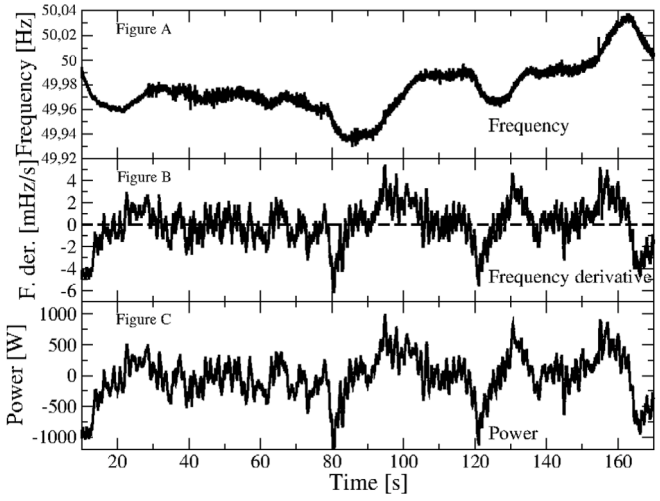


Fig. 12. Figure A shows the grid frequency during a 165 s window. Figure B shows the estimated frequency derivative and Figure C shows the power response of the system while connected to the Nordic grid.

In this system a 2 kHz sampling frequency was chosen for the derivative estimation because a maximum frequency derivative of 35 mHz/s was expected during normal operation.

Then, the system was grid-connected and its operation set as intended, i.e. real voltages were used and the controller was set to generate or regenerate power from the grid depending on the derivative of the frequency. In Fig. 12 the estimated grid frequency and derivative can be seen for a test run of roughly 160 s. The inertia constant K_{SI} was set to 200000, to be able to see a significant power output at low ROCOF. As can be seen, the maximal frequency derivative in the signal is about ±5 mHz/s, giving rise to an active power of around ±1000 W . It can be seen that the derivative follows the frequency signal closely with a small delay and with some noise level. The sign of the power is taken from the SI-systems perspective so a negative value means power is directed from the SI-cabinet to the grid.

At 80 s a larger frequency dip occurred on the grid, as seen as the largest value on the derivative in the window shown; where the system responded with a higher power output. The event is shown zoomed in Fig. 13.

5. Conclusion

A new digital implementation of calculating the grid frequency derivative in real-time using an FPGA and a Savitzky-Golay filter was presented in this paper. The implementation was both simulated in Matlab Simulink and implemented in a National Instruments FPGA controlling a 15 kW Semikron three-phase inverter connected with a supercapacitor connected to the DC-bus. It can be concluded that the grid frequency and frequency derivative can be calculated in real-time and act as a control signal to the converter and present synthetic inertia to the power system. The settling time of the FLL can be decided depending on the criteria of SNR and the expected rate of change of frequency. Depending on the amount of available inertia in the grid and grid frequency SNR, the appropriate window length, filter and loop frequency has to be chosen for the frequency derivative estimator to minimise the noise. It can also be concluded that if the loop frequency is set too low, the phase delay from the FIR filter will result in the total phase delay getting closer to 180°, and the system will be unstable. The system performs well in emulating inertia and could be used as a tool by Transmission System Operators to enhance the system stability.

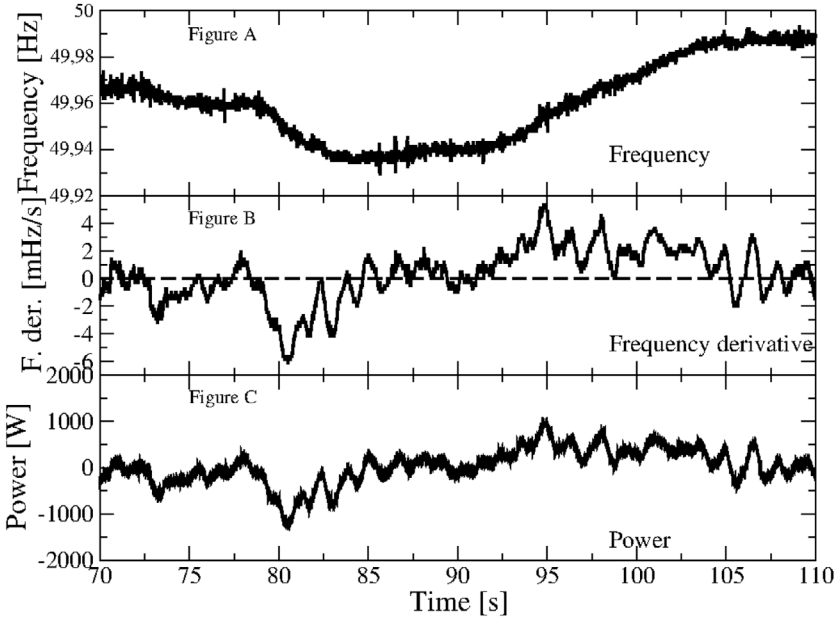


Fig. 13. A Shows the grid frequency during an unbalance in the grid. Figure B shows the estimated grid frequency derivative during the unbalance and Figure C shows the power response of the system while grid connected during the frequency dip.

CRediT authorship contribution statement

Martin Fregelius: Writing, Conceptualization, Software, Validation. **Urban Lundin:** Supervision, Review, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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